

1a. Introduction

Jin Hong

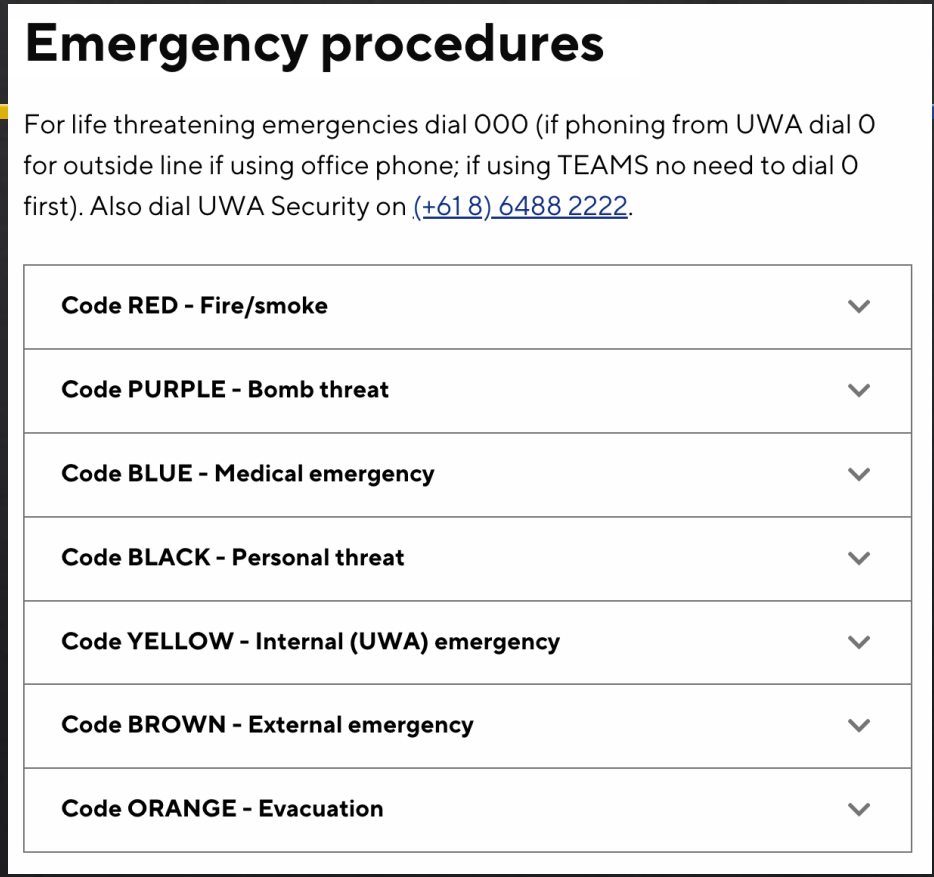
jin.hong@uwa.edu.au

1a. Introduction: outline

- Emergency
- Team
- Location
- Schedule
- Assessments
- Remember

Emergency

- General emergency: call campus security 24/7 at
  (08) 6488 2222
- In super emergency: call emergency at 000
- In all buildings, we have an emergency procedure such as this picture ->
  - Please take a time and read it.
- For more details, please have a read through our emergency procedure for various potential incidents
  - https://www.uwa.edu.au/about-us/location-and-campuses/campus-services/safety/hazards-incidents-and-emergencies/emergency-procedures

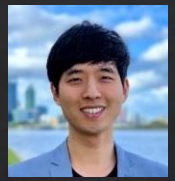


Emergency procedures

For life threatening emergencies dial 000 (if phoning from UWA dial 0 for outside line if using office phone; if using TEAMS no need to dial 0 first). Also dial UWA Security on (+61 8) 6488 2222.

- **Code RED - Fire/smoke**
- **Code PURPLE - Bomb threat**
- **Code BLUE - Medical emergency**
- **Code BLACK - Personal threat**
- **Code YELLOW - Internal (UWA) emergency**
- **Code BROWN - External emergency**
- **Code ORANGE - Evacuation**

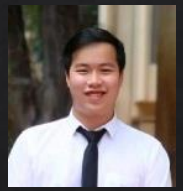
Team



Jin Hong
Unit coordinator
Room CS1.10
jin.hong@uwa.edu.au

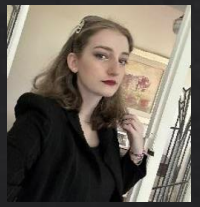


Larry Huynh
Senior Lab Facilitator



Lab: Tue

Alton Wong
Lab facilitator



Lab: Wed

Ari Carter
Lab facilitator



Teaching Operations (team)
Admin team
Room Main reception
teachingops-team2@uwa.edu.au

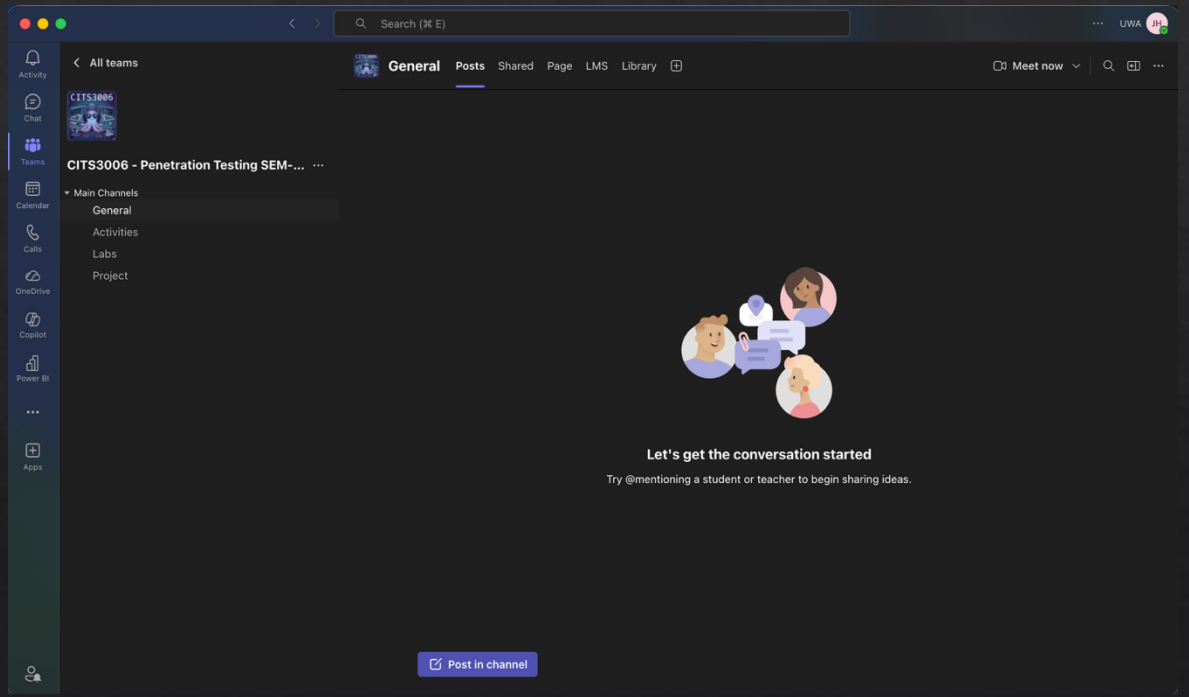
Location

- Lecture
  - Venue: Woolnough Lecture Theatre (225.107)
  - Time: Mondays 1pm - 3pm
- Labs
  - Lab 1: EZONE North (Tuesdays 2 - 4pm)
  - Lab 2: CS lab G11 (Wednesdays 11 - 1pm)
- Consultation
  - Office at Computer Science building room 1.10

Changes will be announced (if any)

Location

- Feel free to ask questions on teams anytime
  - Someone will answer you in their next availability
- You are welcome to join all discussions



Course overview: Term 3

Labs start first week!

Week	Lecture	Lab	Assessments
1	Ethics + Reconnaissance	Lab 0: Setup and Linux	
2	Network Exploits + Malware	Lab 1: Network Exploits	Project out
3	More Malware	Lab 2: Malware	
4	Hardware Security		Lab Quiz 1 + Demo
5	Reverse Engineering	Lab 3: Reverse Engineering	
6	Privilege Escalation	Lab 4: Privilege Escalation	

Course overview: Term 4

Week	Lecture	Lab	Assessments
7	Command and Control		Lab Quiz 2 + Demo
8	Web Security	Lab 5: Web Security	
9	Active Directory	Lab 6: Active Directory	
10	Public holiday on Monday		Lab Quiz 3 + Demo
11	AI in Offensive Security		Project CTF
12	Guest Lectures and/or Revision		Project presentation

Assessments

Assessment Item	When	Covers	Worth (total)
Lab Quizzes	Weeks 4, 7, 10	Materials up to the assessment week (excluding), and also excluding previously assessed weeks.	60% (20% each)
Group Project	Week 2	Various	40%
Study tour research project	over the semester + semester break (4 weeks)	Various	prepares you for honours/masters research

Please note, all dates are tentative and subject to change!

How does the lab quiz work?

- We will be using two formats for the lab quizzes:
  - Complete given tasks over the set amount of time. and do the demo in the lab
  - Complete given tasks and demo during the scheduled lab
- You cannot receive mark if your attendance to the demo is not confirmed by the lab facilitator for both format.
  - i.e., you must be supervised/invigilated by the lab facilitator to receive marks.
  - Attendance is required for all F2F students (unless approved previously e.g., special consideration).
- Missed a lab quiz with an approved EVA? The deferred sitting is a 1-hour, closed-book test unless you're told otherwise in writing.

How does the lab quiz work?

- Demo will be interview-style with the lab facilitator/UC.
- If the demo is the whole lab (i.e., lab quiz completed at another time)
  - You will be formed into a group of 3
  - Each group will have 30 mins to do the interview with the marker
  - Full demo is expected, as instructed by the marker
- If the demo is after the lab quiz in the lab (i.e., 1 hour lab quiz and 1 hour demo)
  - You will be formed into a group of 3
  - Each group will have 15 mins to do the interview with the marker
  - Demo of one solution/method/implementation selected by the marker
- Feedback will be provided to the demonstrator.

Group Project

- Details to be confirmed but this year I am thinking:
  - Your group is a team of cybersecurity experts creating CTF challenges for various CTF events
  - Task 0: You are in a group (week 2)
  - Task 1: populate CTF challenges meeting project requirements (weeks 3 – 10)
  - Task 2: participate in the CITS3006 CTF (week 11)
  - Task 3: present your findings (week 12)

(Optional) Research project

- This year you have an optional study tour to Vietnam (16 Nov – 13 Dec) under the NCP grant scheme – 15 funded spots available
- You don't have to do this – purely optional for enthusiasts
- If interested, contact me directly via email with the header:

  "CITS3006 study tour"

EOI due end of the week!

- There are some strict eligibility rules for the NCP grant:
  - You must be an Australian citizen
  - You have not received the NCP grant previously
  - You must be aged 18 or more
  - Contact me to find about the full details
- If you don't meet the eligibility, but still would like to participate, you may be eligible for UWA Global Learning grant (funding up to $1000) – but you will have to cover the rest of the trip cost.

(Optional) Research project

- You will be assigned in a group of 4~5 students, and a mentor to assist with the research.
- This will be a good practice for your honours research or Masters research project (if you are in BACS or thinking of doing Masters).
  - Such students are given preference
- We'll have regular meetings (weekly) to discuss about the research project and make progress throughout the semester.
- Goal is to publish a paper with accompanying software/tool.
  - One way to fast-track your honours/masters research project

Prerequisite

- Prerequisites: CITS2002 or CITS2005
- Recommended: CITS2006 and CITS3002
- This unit requires the student to be knowledgeable in various aspects of CS.
- This unit expects students to be knowledgeable in various programming languages such as Java, Python and C (or be able to quickly catch up as necessary).
- You should also be familiar with basic discrete mathematics and number theory.
- I highly recommend taking this unit in third year after taking various CITS units (including ones listed above) for preparations.

Please note

- This unit is/does NOT:
  - cover a comprehensive penetration testing techniques
    - Pentesting and cybersecurity is a broad discipline!
  - about all of the latest and greatest attacks
    - Covering selected but important fundamental themes
    - Read online sources instead to keep up to date
  - cover ethical, legal or economic issues
    - We will touch on ethical issues briefly later, but not focus on them

Please note

- Lectures provide **theoretical** and **conceptual** understandings of the topics presented
  - But we will also do some practicals in the lectures
  - So, you should bring your laptop if you want to try them in class as well
- Labs and project provide **practical skills** of the topics presented
- The contents in Lectures, labs and project will be related, but they are all **INDEPENDENT** learning materials
  - i.e., you will be learning new things in each lecture, lab and project.

FAQ

- Why is lab quiz timed?
  - Being able to solve problems in a reasonable time is a key ability in any roles. We have more flexible assessment – group project that runs over multiple weeks.
  - Also, assessment contents are set such that it is reasonable to solve within the given timeframe.
  - You will go through many other assessment methods in your degree, this is just one of them.

- I was told to use chatGPT to prepare for lab quizzes, is this true? why?
  - This is a misconception. You can use chatGPT and alike tools in most, if not all, assessments in CITS3006 to your best ability, but you'll find it quickly that all tasks given are not fully solvable by chatGPT (and alike). We encourage using any and every tool you'll find yourself helpful/useful, including chatGPT, to support your learning, as well as in assessments as it seem fit.
  - We found some students solve all assessments without the use of chatGPT, while some benefited using chatGPT for certain tasks. It is totally up to you if you wish to use it or not.

FAQ

- I feel like the instructions in the assessments are too vague, why?
  - It leaves sufficient room for exploration. Also, in any software development/testing environments, you will never have perfect requirements. It is up to you to consult with the client to check that the requirements are met or make reasonable justifications that would be acceptable by the client.
  - Same in the assessments, you are also told to discuss with me if any confusions arise, and reasonable and practical justifications and assumptions are accepted for design choices of your solutions, especially in the timed assessments.

FAQ

- Why don't we learn about tool X and tool Y?
  - I understand some tools are the preferred tool in certain industries. But the goal of the unit is not to train you to use those tools, but focus on the fundamental aspects of the topics covered.
  - 10 years ago, none of those tools existed (or most).
  - 10 years later, you probably be using some other tools.
  - However, the fundamentals don't change that quickly
    - E.g., thinking about algorithms, data structures etc.

Any other questions?

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